# Exercise Sheet 20

The conventions are those of Lecture 20: $\mathfrak{gl}_k$ is the upper-left block, $Z_k = E_{11} + \cdots + E_{kk}$, and $\mu \prec \lambda$ means interlacing. We use the hook-content formula from the Lecture 18 sheet,

$$\dim \mathbb{S}_\lambda(\mathbb{C}^n) = \prod_b \frac{n + c(b)}{h(b)},$$

where $b$ runs over the boxes of $\lambda$, with content $c(b)$ and hook length $h(b)$. We also use the dimension $\frac{1}{2}(p+1)(q+1)(p+q+2)$ of the irreducible $\mathfrak{sl}_3$-module with Dynkin labels $(p, q)$, from Lecture 17. Exercises 20.1–20.3 are calculations. Exercises 20.4–20.6 compare branching with the symmetric groups and study operators on Gelfand–Tsetlin lines.

**Exercise 20.1 (Branching of $\mathbb{S}_{(2,1)}(\mathbb{C}^4)$).** Let $V = \mathbb{S}_{(2,1)}(\mathbb{C}^4)$.

(i) List the summands of $V$ over $\mathfrak{gl}_3$, their dimensions and the eigenvalues of $E_{44}$. Check the total against the hook-content formula.

(ii) Restrict further to $\mathfrak{gl}_2$. Find all multiplicities and check the dimension.

(iii) Decompose $V$ over $\mathfrak{sl}_3$ and over $\mathfrak{sl}_2$. Which restriction is multiplicity-free?

**Exercise 20.2 (Dimension counts by branching).** Let $k \in \mathbb{Z}_{\geq 0}$.

(i) Show that $\dim \mathbb{S}_{(k)}(\mathbb{C}^n) = \sum_{j=0}^{k} \dim \mathbb{S}_{(j)}(\mathbb{C}^{n-1})$ and deduce $\dim \mathbb{S}_{(k)}(\mathbb{C}^n) = \binom{n+k-1}{k}$.

(ii) For $1 \leq k \leq n$, show that branching of $\mathbb{S}_{(1^k)}(\mathbb{C}^n)$ gives Pascal's rule for $\binom{n}{k}$.

(iii) Show that the Gelfand–Tsetlin lines of $\mathbb{S}_{(k)}(\mathbb{C}^n)$ are the lines spanned by monomials in $\mathrm{Sym}^k(\mathbb{C}^n)$.

**Exercise 20.3 (The sixty-four-dimensional $\mathfrak{sl}_4$ module).** Let $\lambda = (3, 2, 1, 0)$.

(i) Check $\dim \mathbb{S}_\lambda(\mathbb{C}^4) = 64$ by the hook-content formula.

(ii) List the summands over $\mathfrak{gl}_3$ with their $\mathfrak{sl}_3$ Dynkin labels and dimensions. Which $\mathfrak{sl}_3$-module is repeated?

(iii) Show that $t_4 = Z_3 - 3E_{44}$ separates the repeated copies.

**Exercise 20.4 (Comparison with symmetric-group branching).** Restriction from $S_d$ to $S_{d-1}$ sends $V^\lambda$ to the sum of $V^\mu$ over the partitions $\mu$ obtained by removing one box.

(i) Let $\lambda$ have at most $n-1$ parts. Prove that $\{\mu \prec \lambda : |\mu| = |\lambda| - 1\}$ is the set of partitions obtained from $\lambda$ by removing one box.

(ii) Compare the restrictions of $V^{(2,1)}$ to $S_2$ and of $\mathbb{S}_{(2,1)}(\mathbb{C}^3)$ to $\mathfrak{gl}_2$.

(iii) Explain by dimensions why the $\mathfrak{gl}_n$ rule cannot be "remove one box".

**Exercise 20.5 (Operators that change one row).** Let $V = \mathbb{S}_\lambda(\mathbb{C}^n)$ and let $k$ be an integer with $1 \leq k \leq n-1$. Let $x \in \mathfrak{gl}_{k+1}$ commute with $\mathfrak{gl}_{k-1}$. For $k = 1$ there is no commutation condition.

(i) Prove that $xL_\Lambda \subseteq \bigoplus L_{\Lambda'}$, the sum over patterns $\Lambda'$ that agree with $\Lambda$ except possibly in row $k$.

(ii) Show that $E_{k,k+1}$ and $E_{k+1,k}$ satisfy the hypothesis. Show that $E_{k,k+1}$ raises $|\lambda^{(k)}|$ by one.

**Exercise 20.6 (A Gelfand–Tsetlin matrix in $\mathfrak{sl}_3$).** Use the following Gelfand–Tsetlin basis of $\mathfrak{sl}_3$ from the lecture:

$$E_{13},\ E_{23},\ E_{12},\ h,\ E_{21},\ t,\ E_{32},\ E_{31},$$

where $h = E_{11} - E_{22}$ and $t = E_{11} + E_{22} - 2E_{33}$.

(i) Compute $\mathrm{ad}(E_{23})$ on all eight vectors. Check the rule of Exercise 20.5.

(ii) Compute $\mathrm{ad}(E_{32})E_{23}$ in this basis. Find a product of two matrix entries that does not depend on the normalisation.

(iii) Prove directly, using its decomposition in $h$ and $t$, that $E_{22} - E_{33}$ lies in neither Gelfand–Tsetlin line of zero weight.